# What does it look like to have Somya Patnaik on the team?

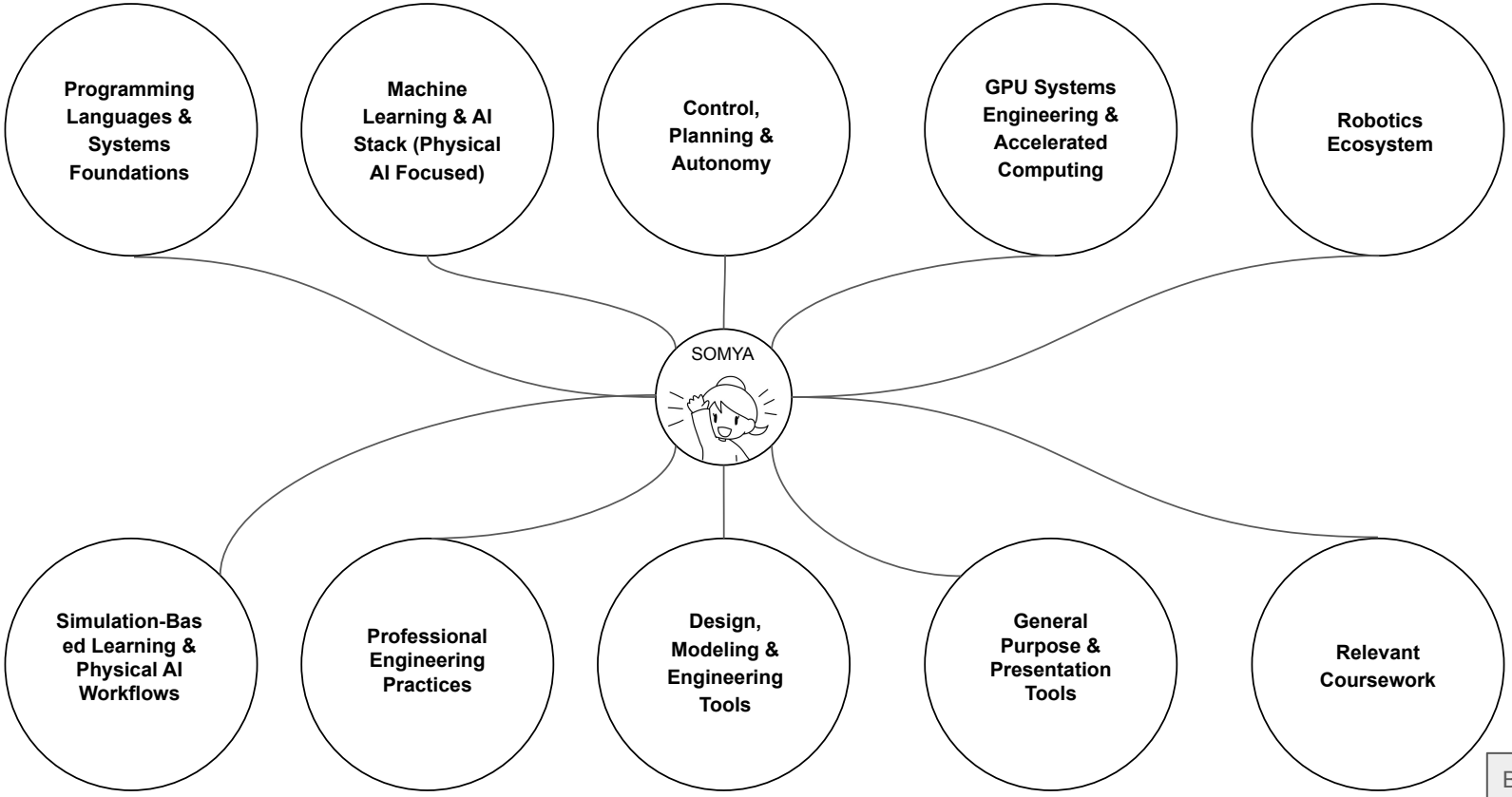
Click to find out

Skill Map

Onboarding Impact

Random Information

# Click on a domain to view the detailed techstack

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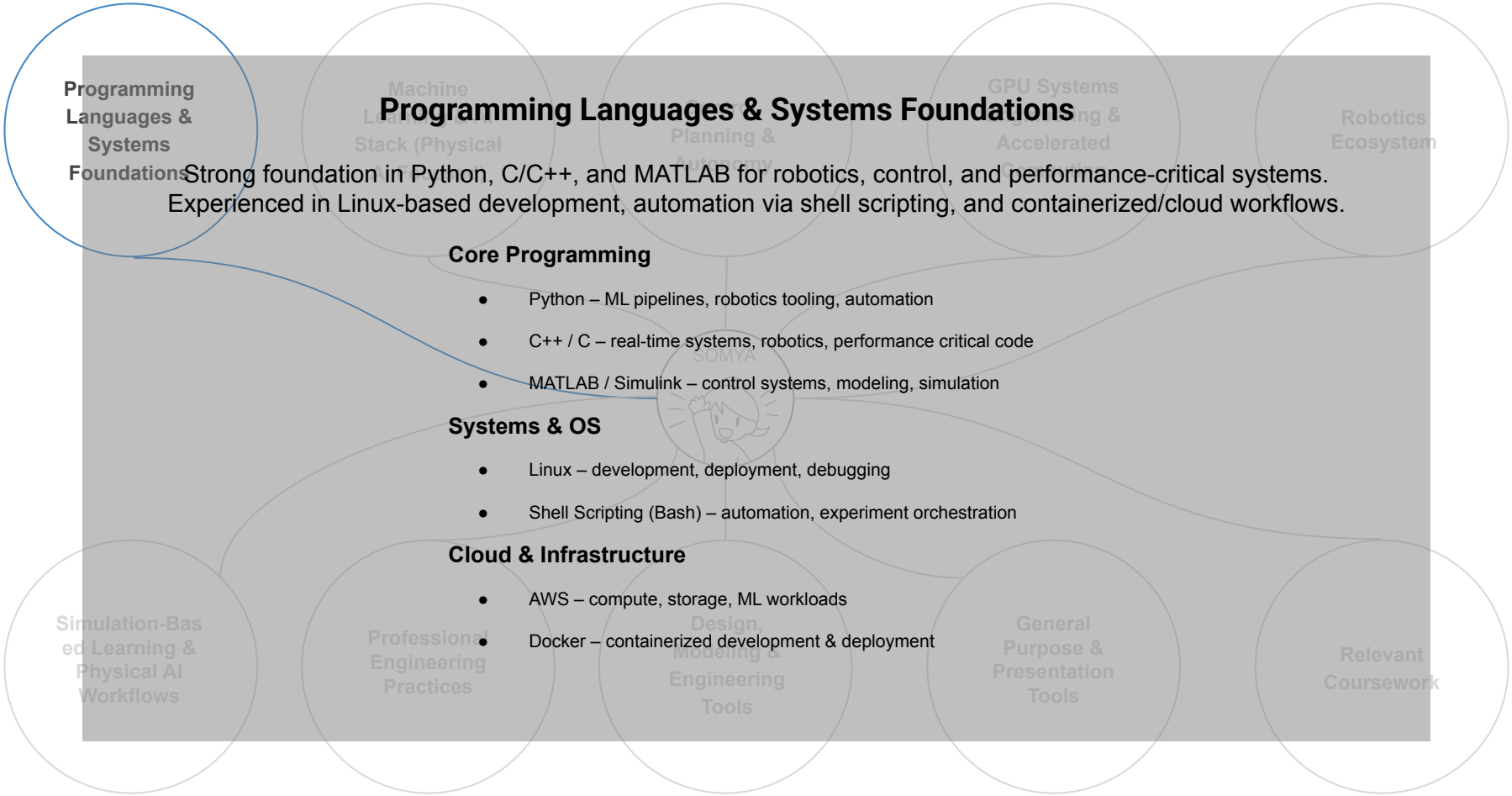


# Programming Languages & Systems Foundations

Strong foundation in Python, C/C++, and MATLAB for robotics, control, and performance-critical systems. Experienced in Linux-based development, automation via shell scripting, and containerized/cloud workflows.

## Core Programming

- Python – ML pipelines, robotics tooling, automation
- C++ / C – real-time systems, robotics, performance critical code
- MATLAB / Simulink – control systems, modeling, simulation

## Systems & OS

- Linux – development, deployment, debugging
- Shell Scripting (Bash) – automation, experiment orchestration

## Cloud & Infrastructure

- AWS – compute, storage, ML workloads
- Docker – containerized development & deployment

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# Machine Learning & AI Stack (Physical AI Focused)

Applied machine learning with a focus on embodied and physical systems rather than standalone models. Experience across vision, numerical computing, GPU acceleration, and optimized AI deployment.

### ML Frameworks

- PyTorch – model development, research workflows
- TensorFlow – training & deployment pipelines
- Scikit-learn – classical ML, prototyping

### Scientific Computing

- NumPy, SciPy – numerical methods, optimization
- Pandas – data pipelines & analysis

### Computer Vision & Perception

- OpenCV – vision pipelines, feature extraction
- Vision-based Mapping – camera-centric perception

### Advanced AI

- CUDA – GPU-accelerated ML & simulation
- TensorRT – optimized inference
- LLMs – model integration, tool-based AI workflows

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# Control, Planning & Autonomy

Deep grounding in kinematics, control theory, and motion planning for real-world autonomous systems.
Experience integrating perception, planning, and control into closed-loop autonomy pipelines.

## Kinematics & Dynamics

- Forward & Inverse Kinematics (FK / IK)
- Jacobians & differential motion
- Rigid-body dynamics

## Control Theory

- Model Predictive Control (MPC)
- Optimal & feedback control
- Constraint-based control

## Motion Planning

- Sampling-based planners(RRT, PRM)
- Optimization-based planners
- Trajectory generation & smoothing

## Autonomous Systems

- AMR Navigation
- Perception–planning–control integration
- Sensor-driven autonomy loops

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# GPU Systems Engineering & Accelerated Computing

Hands-on experience with CUDA-based parallel computing and GPU performance optimization.
Focused on profiling, memory efficiency, and GPU-accelerated physics for high-performance systems.

### CUDA & Parallel Computing

- CUDA Programming (Kernels, Shared memory, Streams & CUDA Graphs)
- Parallel algorithms
- CPU–GPU coordination

### Profiling & Optimization

- Nsight Systems
- Performance bottleneck analysis
- Memory & latency optimization

### GPU Physics & Simulation

- PhysX
- Flex
- GPU-accelerated physics concepts

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# Robotics Ecosystem

Proficient in modern robotics middleware and system integration using ROS 2 and MoveIt 2.
Experience with perception, SLAM, and multi-sensor calibration in robotics pipelines.

### Robotics Middleware

- ROS 2
- MoveIt 2

### NVIDIA Robotics SDKs

- Isaac Perceptor
- Isaac Manipulator
- cuRobo
- GROOT

### Perception & Mapping

- SLAM
- Point cloud processing
- Camera & LiDAR calibration

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# Simulation-Based Learning & Physical AI Workflows

Extensive use of physics-based simulation for robotics development and validation.
Focused on digital twins, synthetic data generation, and simulation-driven learning workflows.

### Simulation Platforms

- Gazebo
- Isaac Sim
- Isaac Lab

### NVIDIA Omniverse Ecosystem

- Omniverse Replicator
- USD (Universal Scene Description)
- Synthetic data generation
- Domain randomization

### Simulation-Based Learning

- Simulation-to-real workflows
- Physics-informed learning
- Simulation-based reinforcement learning

### Physical AI Workflows

- GPU-accelerated robotics pipelines
- Perception, planning, and control integration
- Synthetic data-driven training

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# Professional Engineering Practices

Strong software engineering discipline with version control, CI/CD, and reproducible workflows.
Experienced in experiment automation, documentation, and collaborative development.

### Development Workflow

- Git – version control, collaboration
- CI/CD – automated testing & deployment
- Dockerized workflows

### Experimentation & Automation

- Python + Bash experiment pipelines
- Reproducible research workflows
- Logging & benchmarking

### Documentation & Collaboration

- Technical documentation
- Design & research reports
- Open-source contributions

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# Design, Modeling & Engineering Tools

Background in mechanical design, structural analysis, and CAD for cyber-physical systems.
Enables close collaboration between software, simulation, and hardware design teams.

**Mechanical & Structural**

- ANSYS
- Abaqus
- HyperMesh

**CAD & Product Design**

- SOLIDWORKS
- AutoDesk Fusion 360
- GDT (Geometric Dimensioning & Tolerancing)

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# General Purpose & Presentation Tools

Skilled in visual design, video editing, and professional documentation.
Supports effective communication of technical concepts, demos, and research results.

- Adobe Photoshop
- SONY VEGAS
- Microsoft Office
- Google Workspace

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# Relevant Coursework

Expertise in mathematics, control, optimization, and physics-based system modeling.

**Computer Science**

Embedded Systems and RTOS, Soft Computing, DSA, Problem Solving using C++, Python, Linux and Shell scripting

**ML/AI**

Fundamentals of Machine Learning, RL and learning-based control, AI and Data Analytics, Advanced Machine Learning, Computer Vision

**Control Systems and Electronics**

Digital System Design, Sensors and PLC, Microcontroller-based system design, Linear Control Theory, Digital Signal Processing

**Mechanical**

Automated Manufacturing Systems, Drives and Controls, Hydraulic and Pneumatic Systems, Theory of Machines, Micro Electro Mechanical Systems, Multibody Dynamics, International Factory planning

**Robotics**

Path Planning, Mobile Robots, Advanced Robot Kinematics and Dynamics, Soft Robotics, Mechatronics and Controls, Robotics Sensor Systems, Simulation of Robotic Systems, Design of Machine Elements

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# What I would try to accomplish in the first 15 days

**Days 1- 5: Understanding and alignment**

Goal: Basic setup, observe and plan

- Understand the overall product stack and how my assigned tasks fits into the ecosystem.
- Setup the developer environment.
- Explore existing codebase and benchmarks to understand the development paradigm followed by the team.
- Interact with the team to understand ongoing dev-plan and priorities.

**Days 6- 10: Deeper immersion**

Goal: Build system level intuition and understand technical context.

- Explore the core aspects of my role by reproducing existing work.
- Identify the gaps between development and production requirements.
- Understand all the deployment aspects and data maintenance protocols.
- Interact with the project management team to understand the tools they use and get a sense of their expectations.

**Days 10- 15: Contribution and optimization**

Goal: Make impactful but low risk contribution

- Make an initial contribution such as a bug-fix, tooling enhancement or documentation.
- Discuss and propose a measurable optimization.
- Maintain strong documentation from very beginning to ensure efficient knowledge sharing with the team.

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# Few things the team might find interesting

- Published in 2 peer-reviewed journals (Elsevier). I am looking for my next review domain so I would love to go through research articles and summarize it for them.
- Top 1% in Project Euler.
- Formula 1 enthusiast.
- My commit messages are awesome.
- 3blue1brown nerd.
- I show appreciation with cakes :)

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